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INTERNATIONAL UNIVERSITY

NAME: UKASH IRFAN **SAP ID: 74117**

Node 1: Single Node Creation

PROGRAM CODE

```
#include<iostream>
using namespace std;
struct Node{
        int data;
        Node*next;
};
int main(){
        Node*head=new Node;
        head->data=10;
        head->next=NULL;
        cout<<head->data;
        return 0;
}
```

OUTPUT

```
10
```

Node 2: Two Linked Nodes

PROGRAM CODE

```
#include<iostream>
using namespace std;
struct Node{
        int data;
        Node*next;
};
int main(){
        Node*head=new Node;
        Node*second=new Node;
        head->data=10;
        head->next=second;
        second->data=20;
        second->next=NULL;
        cout<<head->data<<endl;
        cout<<second->data;
        return 0;
}
```

OUTPUT

```
10
20
```

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Node 3: Three Linked Nodes

PROGRAM CODE

```
#include<iostream>
using namespace std;
struct Node{
        int data;
        Node*next;
};
int main(){
        Node*head=new Node;
        Node*second=new Node;
        Node*third=new Node;
        head->data=10;
        head->next=second;
        second->data=20;
        second->next=NULL;
        third->data=30;
        third->next=NULL;
        cout<<head->data<<endl;
        cout<<second->data<<endl;
        cout<<third->data;
        return 0;
}
```

OUTPUT

```
10
20
30
```

Node 4: Traversal with While Loop

PROGRAM CODE

```
#include<iostream>
using namespace std;
struct Node{
        int data;
        Node*next;
};
int main(){
        Node*head=new Node;
        Node*second=new Node;
        head->data=10;
        head->next=second;
        second->data=20;
        second->next=NULL;
        Node*current=head;
        while(current != NULL){
                cout<<current->data<<endl;
                current=current->next;
        }
        return 0;
}
```

OUTPUT

```
10
20
```

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Node 5: LinkedList Class Shell Construction

PROGRAM CODE

```
#include<iostream>
using namespace std;
struct Node{
        int data;
        Node*next;
};
class linkedlist{
        private:
                Node*head;
        public:
                linkedlist(){
                        head=NULL;
                }
                void display(){
                        Node*current=head;
                        while(current != NULL){
                                cout<<current->data<<endl;
                                current=current->next;
                        }
                        cout<<endl;
                }
};
int main(){
        linkedlist list;
        return 0;
}
```

OUTPUT

```
(Program executed with no output)
```

Node 6: Dynamic N-Node Input Construction

PROGRAM CODE

```
#include<iostream>
using namespace std;
struct Node{
        int data;
        Node*next;
};
int main(){
        Node*head=NULL;
        Node*current=NULL;
        int n;
        cout<<"Enter a number How many Node you want : ";
        cin>>n;
        for(int i=1; i<=n; i++){
                Node*newNode= new Node;
                cout<<"Enter data for "<<i<<" Node : ";
                cin>>newNode->data;
                newNode->next=NULL;
                if (head==NULL){
                        head=newNode;
                        current=newNode;
                }
                else{
                        current->next=newNode;
                        current=newNode;
                }
        }
        cout<<"Linked list : "<<endl;
        current=head;
        while(current != NULL){
                cout<<current->data<<endl;
```

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Node 7: LinkedList Class with Append Method

PROGRAM CODE

```
#include <iostream>
using namespace std;
struct Node
{
    int data;
    Node* next;
};
class LinkedList
{
private:
    Node* head;
public:
    LinkedList()
    {
        head = NULL;
    }
    void append(int value)
    {
        Node*newNode = new Node;
        newNode->data = value;
        newNode->next = NULL;
        if (head == NULL)
        {
            head = newNode;
        }
        else
        {
            Node* current = head;
            while (current->next != NULL)
            {
                current = current->next;
            }
            current->next = newNode;
        }
    }
    void display()
    {
        Node* current = head;
        while (current != NULL)
        {
            cout << current->data <<endl;
            current = current->next;
        }
    }
};
int main()
{
    LinkedList list;
    list.append(10);
    list.append(20);
    list.append(30);
    list.append(40);
    cout << "Linked List : "<<endl;
    list.append(50);
    list.display();
    return 0;
}
```

OUTPUT

```
Linked List :
10
20
30
40
50
```

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Node 8: LinkedList Insertion at Specific Position

PROGRAM CODE

```
#include <iostream>
using namespace std;
struct Node
{
    int data;
    Node* next;
};
class LinkedList
{
private:
    Node* head;
public:
    LinkedList()
    {
        head = NULL;
    }
    void append(int value)
    {
        Node*newNode = new Node;
        newNode->data = value;
        newNode->next = NULL;
        if (head == NULL)
        {
            head = newNode;
        }
        else
        {
            Node* current = head;
            while (current->next != NULL)
            {
                current = current->next;
            }
            current->next = newNode;
        }
    }
    void insert(int position, int value)
    {
        Node* newNode = new Node;
        newNode->data = value;
        if (position == 1)
        {
            newNode->next = head;
            head = newNode;
        }
        else
        {
            Node* current = head;
            for (int i = 1; i < position - 1; i++)
            {
                current = current->next;
            }
            newNode->next = current->next;
            current->next = newNode;
        }
    }
    void display()
    {
        Node* current = head;
        while (current != NULL)
        {
            cout << current->data <<endl;
            current = current->next;
        }
    }
};
int main()
```

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Node 9: Complete LinkedList with User Input, Insertion & Deletion

PROGRAM CODE

```
#include <iostream>
using namespace std;
struct Node
{
    int data;
    Node* next;
};
class LinkedList
{
private:
    Node* head;
public:
    LinkedList()
    {
        head = NULL;
    }
   void user(){
        Node*current=NULL;
        int n;
        cout<<"Enter a number How many Node you want : ";
        cin>>n;
        for(int i=1; i<=n; i++){
                Node*newNode=new Node;
                cout<<"Enter data for "<<i<<" Node : ";
                cin>>newNode->data;
                newNode->next=NULL;
                if (head==NULL){
                        head=newNode;
                        current=newNode;
                }
                else{
                        current->next=newNode;
                        current=newNode;
                }
}
                cout<<"Orignal Linked list : "<<endl;
                current=head;
                while(current != NULL){
                        cout<<current->data<<endl;
                        current=current->next;
                        }
}
    void insert(int position, int value)
    {
        Node* newNode = new Node;
        newNode->data = value;
        if (position == 1)
        {
            newNode->next = head;
            head = newNode;
        }
        else
        {
            Node* current = head;
            for (int i = 1; i < position - 1; i++)
            {
                current = current->next;
            }
            newNode->next = current->next;
            current->next = newNode;
        }
    }
    void remove(int position){
        if(position==1){
                Node* temp = head;
```

```
                head = head->next;
                        delete temp;
```